

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I Mangali Veera Naga Sai (192425253) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Veera Naga Sai

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

AI-Based Medical Diagnosis

AI-Based Medical Diagnosis — represented by AI-powered healthcare systems such as Google DeepMind, IBM Watson Health, Aidoc, PathAI, and Viz.ai. This application belongs to the Healthcare domain of Artificial Intelligence and is used to assist doctors in detecting diseases, analyzing medical images, interpreting patient data, and supporting clinical decision-making for faster and more accurate diagnosis.

1. Domain Selection

Domain: Healthcare. This report focuses specifically on **AI-Based Medical Diagnosis**, where Artificial Intelligence systems analyze medical images, patient records, laboratory reports, and clinical data to assist healthcare professionals in detecting diseases, supporting diagnosis, and improving treatment decisions with greater speed and accuracy.

2. Introduction and Background

AI-Based Medical Diagnosis uses Artificial Intelligence to assist healthcare professionals in detecting diseases, analyzing medical images, and interpreting patient data with high accuracy. AI systems can identify patterns in X-rays, CT scans, MRI scans, laboratory reports, and electronic health records that may be difficult or time-consuming for humans to detect. By supporting faster and more reliable diagnosis, AI helps improve treatment planning and patient outcomes.

In **2025–2026**, AI-based medical diagnosis has seen rapid adoption in hospitals, diagnostic centers, and research institutions worldwide. Leading AI platforms such as **Google DeepMind, IBM Watson Health, Aidoc, PathAI, and Viz.ai** are being used to detect conditions such as cancer, stroke, heart disease, and diabetic retinopathy. These systems assist radiologists and physicians by providing accurate diagnostic insights, reducing reporting time, and enabling earlier disease detection, making AI an increasingly important part of modern healthcare.

3. Problem Identification

Healthcare systems face increasing patient numbers, shortages of experienced medical professionals, and the growing complexity of diagnosing diseases. Manual analysis of medical images, laboratory reports, and patient records is often time-consuming and may lead to delayed diagnosis or human error, especially in busy hospitals.

The real-world problem is therefore: **how can diseases be detected quickly and accurately while reducing diagnostic errors and supporting healthcare professionals in making timely treatment decisions?** AI-Based Medical Diagnosis addresses this challenge by analyzing medical data, identifying disease patterns, and providing decision support, enabling faster, more accurate, and efficient healthcare services.

4. Need for AI Solution

Traditional medical diagnosis relies on doctors manually analyzing medical images, laboratory reports, and patient records. While effective, this process can be time-consuming, depends heavily on the experience of healthcare professionals, and may lead to delayed diagnosis or human error, especially when handling large volumes of patient data.

- **Traditional diagnosis:** Manual analysis, time-consuming, and dependent on physician expertise.
- **AI-based diagnosis:** Analyzes medical images and patient data quickly, detects hidden patterns, and provides decision support with high accuracy.
- **AI enables healthcare professionals to diagnose diseases earlier, reduce diagnostic errors, and improve treatment planning**, making healthcare services faster, more efficient, and more reliable.

5. AI Technologies Used

Modern **AI-Based Medical Diagnosis** systems integrate several AI technologies to analyze medical data and support accurate disease detection:

- **Machine Learning (ML)** — Learns patterns from patient records, laboratory results, and medical history to predict diseases and assist diagnosis.
- **Deep Learning (DL)** — Uses neural networks to analyze medical images such as X-rays, CT scans, MRI scans, and pathology slides for detecting abnormalities.
- **Computer Vision** — Processes and interprets medical images to identify diseases such as cancer, fractures, pneumonia, and diabetic retinopathy.
- **Natural Language Processing (NLP)** — Extracts useful information from electronic health records, doctor's notes, and clinical reports to support diagnosis and treatment decisions.
- **Predictive Analytics** — Combines patient data, medical history, and AI models to estimate disease risk, recommend early interventions, and improve clinical decision-making.

6. Working Mechanism

An **AI-Based Medical Diagnosis** system follows the process: **Input** → **Data Processing** → **AI Analysis** → **Diagnosis** → **Feedback**.

- **Input:** Medical images (X-rays, CT scans, MRI scans), laboratory test results, electronic health records, and patient symptoms are collected.
- **Processing:** The data is cleaned, organized, and preprocessed. AI models extract important features from medical images and patient information.
- **AI Analysis:** Machine Learning and Deep Learning models analyze the data, identify disease patterns, compare them with trained datasets, and predict possible medical conditions.
- **Diagnosis:** The AI system provides diagnostic results, highlights abnormal findings, and suggests possible diseases or risk levels for review by healthcare professionals.
- **Feedback:** Doctors verify the AI predictions, make the final diagnosis, and the confirmed results can be used to improve the AI model for future diagnostic accuracy.

7. Data Sources and Processing

Training data for **AI-Based Medical Diagnosis** comes from multiple healthcare sources: (1) **medical imaging datasets** such as X-rays, CT scans, MRI scans, and pathology images; (2) **electronic health records (EHRs)** containing patient history, symptoms, diagnoses, and treatment information; and (3) **laboratory test results** and clinical reports collected from hospitals and healthcare institutions. These datasets are carefully labeled by medical experts to ensure accurate training.

The collected data is preprocessed by removing errors, handling missing values, normalizing images, and extracting important features. Machine Learning and Deep Learning models are then trained to recognize disease patterns and make diagnostic predictions. As more verified patient data becomes available, the AI models are continuously updated and improved to increase diagnostic accuracy and support better clinical decision-making.

8. Real-World Implementation

Verified real-world implementations of **AI-Based Medical Diagnosis** as of **2025–2026** include:

- **Hospitals and Diagnostic Centers:** **Google DeepMind** assists doctors in detecting eye diseases, kidney disorders, and breast cancer through AI-powered medical image analysis.
- **Radiology:** **Aidoc** and **Viz.ai** are used in hospitals to analyze CT scans, MRI scans, and X-rays, helping doctors detect stroke, brain injuries, and other critical conditions more quickly.
- **Pathology:** **PathAI** supports pathologists by analyzing digital pathology slides to improve the detection of cancer and other diseases.
- **Clinical Decision Support:** **IBM Watson Health** has been used to analyze patient records and medical literature to assist healthcare professionals in making informed diagnosis and treatment decisions.

The global **AI in Healthcare** market is expected to grow rapidly over the next decade, driven by increasing adoption of AI technologies in disease diagnosis, medical imaging, personalized treatment, and patient care across hospitals and healthcare organizations worldwide.

9. Impact and Benefits

- ❑ **Improved Accuracy:** AI helps detect diseases more accurately by analyzing medical images, patient records, and clinical data, reducing diagnostic errors.
- ❑ **Faster Diagnosis:** AI processes large amounts of healthcare data in a short time, enabling early disease detection and quicker treatment decisions.
- ❑ **Better Patient Care:** AI assists doctors in developing personalized treatment plans, improving patient outcomes and overall healthcare quality.
- ❑ **Increased Efficiency:** AI reduces the workload of healthcare professionals by automating routine diagnostic tasks, allowing doctors to focus on patient care and complex medical cases.

10. Challenges and Limitations

- Technical: dynamic balance, dexterous grasping of irregular objects, and long-horizon task planning remain unsolved in unstructured settings.
- Economic: current commercial units (for example, Agility's Digit) cost far more than comparable fixed automation, and payback periods are still being proven.
- Safety and standards: industrial safety frameworks such as ISO 10218 must be adapted for robots that share open floor space with humans, rather than working in caged cells.
- Workforce impact: displacement of warehouse and assembly-line labour raises ethical and social questions requiring workforce transition planning.
- Data and privacy: continuous camera-based perception in workplaces raises surveillance and data-governance concerns.

11. Future Scope and Emerging Trends

- **Advanced AI Models:** Future AI systems will use more powerful Machine Learning and Deep Learning models to improve diagnostic accuracy and detect diseases at earlier stages.
- **Personalized Healthcare:** AI will provide personalized treatment recommendations by analyzing a patient's medical history, genetic information, and lifestyle data.
- **Remote Healthcare Integration:** AI-powered diagnostic tools will be increasingly integrated with telemedicine and wearable devices for continuous patient monitoring and remote diagnosis.
- **Wider Adoption:** As AI technology becomes more reliable and affordable, its use is expected to expand across hospitals, clinics, and healthcare centers worldwide, improving access to quality healthcare.

12. Conclusion

AI-Based Medical Diagnosis demonstrates how Artificial Intelligence technologies such as **Machine Learning, Deep Learning, Computer Vision, and Natural Language Processing** can improve the speed and accuracy of disease detection and clinical decision-making. Although challenges such as data privacy, regulatory compliance, and the need for expert validation remain, **2025–2026** has seen widespread adoption of AI-powered diagnostic systems in hospitals and healthcare organizations. With continuous improvements in AI models, medical data availability, and healthcare infrastructure, AI-Based Medical Diagnosis is expected to play a vital role in enabling earlier disease detection, personalized treatment, and better patient outcomes in the future.

13. References

- Google DeepMind, "AI for Healthcare and Medical Diagnosis," 2026.
- [2] IBM, "IBM Watson Health – AI Solutions for Healthcare," 2026.
- [3] Aidoc, "AI-Powered Radiology and Clinical AI Platform," 2026.
- [4] PathAI, "Artificial Intelligence for Digital Pathology," 2026.
- [5] Viz.ai, "AI-Based Stroke Detection and Care Coordination," 2026.
- [6] World Health Organization (WHO), "Ethics and Governance of Artificial Intelligence for Health," 2025.
- [7] National Institutes of Health (NIH), "Artificial Intelligence in Healthcare: Current Applications and Future Directions," 2025.
- [8] Nature Medicine, "Artificial Intelligence in Clinical Diagnosis and Medical Imaging," 2025.
- [9] Grand View Research, "Artificial Intelligence in Healthcare Market Size, Trends, and Forecast (2025–2030)," 2026

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement (60–50%)
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.